Anubha Vyasamudri

(669) 296-9945 | av3329@columbia.edu | www.linkedin.com/in/anubha-vyasamudri/

EDUCATION

Columbia University — New York, NY
Master of Science in Computer Science — Expected Dec 2026
Courses: Analysis of Algorithms, Databases, Machine Learning, Big Data Analytics
Teaching Assistant: Neural Networks and Deep Learning, Machine Learning for Data Science

Indian Institute of Technology Bombay — Mumbai, IN
Bachelor of Technology in Mechanical Engineering, Minor in Artificial Intelligence and Data Science — May 2025
Courses: Programming for Data Science, Probability & Stochastic Processes, Linear Algebra, Multivariable Calculus
Teaching Assistant: Statistical Machine Learning and Data Mining

PROFESSIONAL EXPERIENCE

Kahuna Labs — San Jose, CA
AI Engineer Intern — May 2026 – Aug 2026

- Built severity- and escalation-aware LLM response workflows with cross-team input, tailoring diagnostic questions and follow-ups to client needs; integrated OpenAI and Gemma/vLLM evaluators with live Atlas cases, caching, and concurrent execution.
- Improved production reliability with rate-limit backoff, retries, and isolated HTTP clients, eliminating manual quality checks.
- Built an LLM-as-a-Judge framework for support-ticket next steps with precision/recall and trace-level evaluation results; analyzed errors by severity, complexity, case age, and product to identify response-quality issues in client-support workflows.
- Built GPT-4o/open-weight VLM extraction with Pydantic validation and prompt tuning; passed 23/23 GPT-4o test images.

American Express — Gurugram, IN
Analyst Intern - Global Merchant & Network Services — May 2024 – Jul 2024

- Collaborated cross-functionally to build a Python/SQL ML framework optimizing merchant offers and portfolio ROI.
- Engineered gradient boosting features (75% accuracy); tuned offers for 95% ROI and 82% merchant-value predicted success.
- Applied predictive modeling to estimate offer success probabilities and support data-driven merchant portfolio decisions.

General Aeronautics Pvt Ltd — Bengaluru, IN
Data Science Intern — May 2023 – Jul 2023

- Developed RetinaNet and Faster R-CNN object detection pipelines for drone navigation and agricultural field mapping.
- Built and benchmarked TensorFlow/PyTorch training and inference workflows for object detection on 600+ stitched TIFF images; evaluated detection performance across agricultural field conditions for drone navigation and field mapping.

PROJECT AND RESEARCH EXPERIENCE

Bounded Agent for Security Alert Classification | Python, OpenAI, ReAct, Bandit — New York, NY
Course Project, Columbia University — Oct 2026 – Present

- Implemented ReAct agent control flow with token-budget gates, failure recovery, and deterministic handling of unfinished alerts.
- Built JSONL audit traces and schema-validated reports; co-evaluated 49 security alerts across 3 codebases (75% OWASP recall).

PawAI: Multi-Agent System for Personalized Dog Care — New York, NY
Course Project, Columbia University — Jan 2026 – Mar 2026

- Implemented Flask endpoints and a Streamlit interface for a dog-care assistant, integrating quantized Llama inference with health, nutrition, and training agents; synthesized multi-agent responses with citations and safety-aware health triage.
- Built persistent FAISS retrieval indexes, JSON agent planning, and fallback routing for grounded responses to user questions.

Expression AI: Real-Time De-escalation Engine for Conversational LLMs — New York, NY
Course Project, Columbia University — Mar 2026 – May 2026

- Built a shared training-and-demo package integrating escalation prediction, SHAP-driven prompt adaptation, and Mistral generation; used conversation-level train/test splits across 50K conversations and 340K turns to prevent evaluation leakage.
- Compared XGBoost and BiLSTM models; saved held-out evaluations recorded ROC AUCs of 0.790 and 0.773, respectively.

Uncertainty-Aware Inference: Post-Training Quantization & LLM Calibration. — New York, NY
Course Project, Columbia University - Mentored by IBM Research — Jan 2026 – May 2026

- Profiled GPU kernels and vLLM serving; identified a Mistral GPTQ INT4 configuration with 1.78x FP16 batched throughput.
- Evaluated GPTQ, AWQ, and NF4 quantization across LLMs using ECE, MCE, Brier Score, and out-of-distribution benchmarks.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (Node.js), SQL, C++
AI/LLM: OpenAI API, RAG, FAISS, LangChain, Agent Orchestration, LLM Evaluation, Prompt Engineering, Structured Outputs
Platform & Backend: Flask, FastAPI, REST APIs, Pydantic, Async Processing, Concurrent Programming, Git
Systems: Linux, Docker, NVIDIA Nsight Compute, GPU Profiling, vLLM